

CHAPTER 16

Aqueous solutions

Aqueous solutions are found all around you. The Earth's oceans, rivers and lakes are aqueous solutions. They contain dissolved minerals and gases. The plasma of human blood and the sap of plants are aqueous solutions carrying dissolved nutrients and wastes. Each body cell contains aqueous solutions. Even rain contains small quantities of dissolved gases and other materials.

In fields such as medicine, pharmaceutical manufacturing, and even food preparation, it is very important to know how much of a compound is present in a solution.

In this chapter, you will learn how to determine how much of a substance will dissolve in a particular solvent. You will also explore the nature of aqueous solutions and learn how to calculate the amount of a compound dissolved in a given solution.

Science understanding

- the presence of specific ions in solutions can be identified by observing the colour of the solution, flame tests and observing various chemical reactions, including precipitation and acid–base reactions
- solutions can be classified as saturated, unsaturated or supersaturated; the concentration of a solution is defined as the quantity of solute dissolved in a quantity of solution; this can be represented in a variety of ways, including by the number of moles of the solute per litre of solution (mol L^{-1}) and the mass of the solute per litre of solution (g L^{-1}) or parts per million (ppm)
- the mole concept can be used to calculate the mass of solute, and solution concentrations and volumes involved in a chemical reaction

16.1 Precipitation reactions

A **precipitation reaction** occurs if ions in solution combine to form a new compound that is insoluble in water. The insoluble compound formed in such a reaction is called a **precipitate**.

Precipitation reactions occur naturally in undersea hydrothermal vents. The vents release superheated solutions containing sulfides, which then combine with metal ions to form precipitates of mineral sulfides, creating the chimney-like structures seen in Figure 16.1.1. The areas around these chimneys are biologically rich, often hosting complex communities fuelled by the chemicals dissolved in the vent fluids.

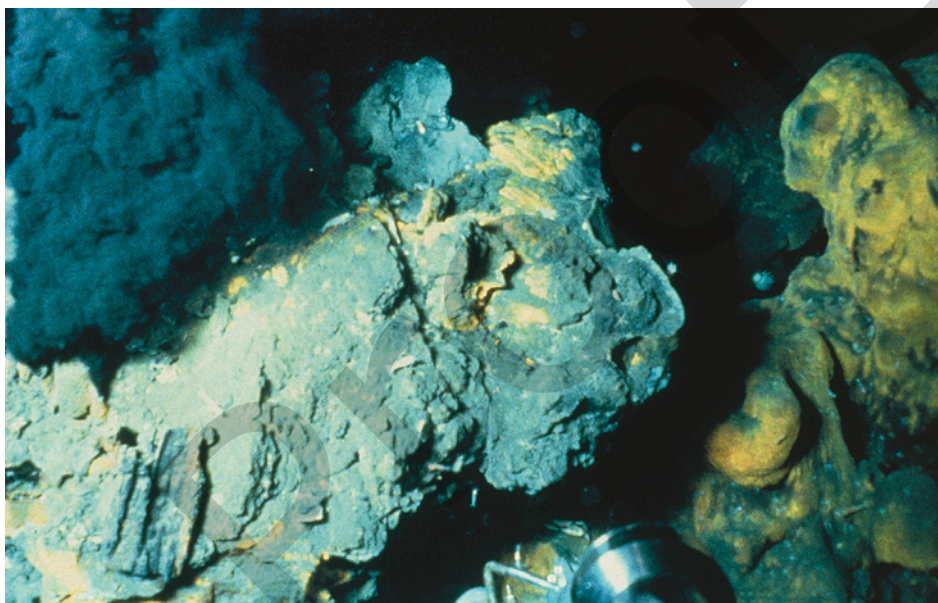


FIGURE 16.1.1 Undersea hydrothermal vents release superheated water containing sulfides, which form precipitates with metal ions.

Precipitation reactions are used to remove minerals from drinking water, to remove heavy metals from wastewater and in the purification plants of reservoirs.

In this section you will look at what takes place during precipitation reactions.

DEDUCING THE IDENTITY OF A PRECIPITATE

When a colourless solution of silver nitrate is mixed with a colourless solution of sodium chloride, a white solid is formed, as shown in Figure 16.1.2. The white solid is the precipitate, but what is its composition? What is the solid compound that forms?

To understand what happens in the reaction between silver nitrate and sodium chloride, you need to identify the ions present in the reactant solutions and how they interact with each other.

- In the silver nitrate solution, there are silver ions (Ag^+) and nitrate ions (NO_3^-).
- In the sodium chloride solution, there are sodium ions (Na^+) and chloride ions (Cl^-).
- When one solution is added to the other, the mixture formed will contain all of the ions.

In both of these solutions, all the ions are moving around independently. As the ions move in the solution, they will collide with one another. If cations (positive ions) and anions (negative ions) collide, they may join together to form a new, insoluble precipitate.

Two new combinations of cations and anions are possible:

- sodium and nitrate ions (to form sodium nitrate)
- silver and chloride ions (to form silver chloride).



FIGURE 16.1.2 Mixing aqueous solutions of sodium chloride and silver nitrate produces a solid, called a precipitate.

Tables 15.5.1 and 15.5.2 on page xxx indicate that sodium nitrate is soluble in water, but that silver chloride is insoluble. Therefore, the precipitate must be silver chloride.

The process for the precipitation reaction between sodium chloride and silver nitrate is shown in Figure 16.1.3. When the hydrated Ag^+ and Cl^- ions come into contact, the attraction between the ions is greater than that of the ions for the water molecules. An ionic lattice of AgCl is formed.

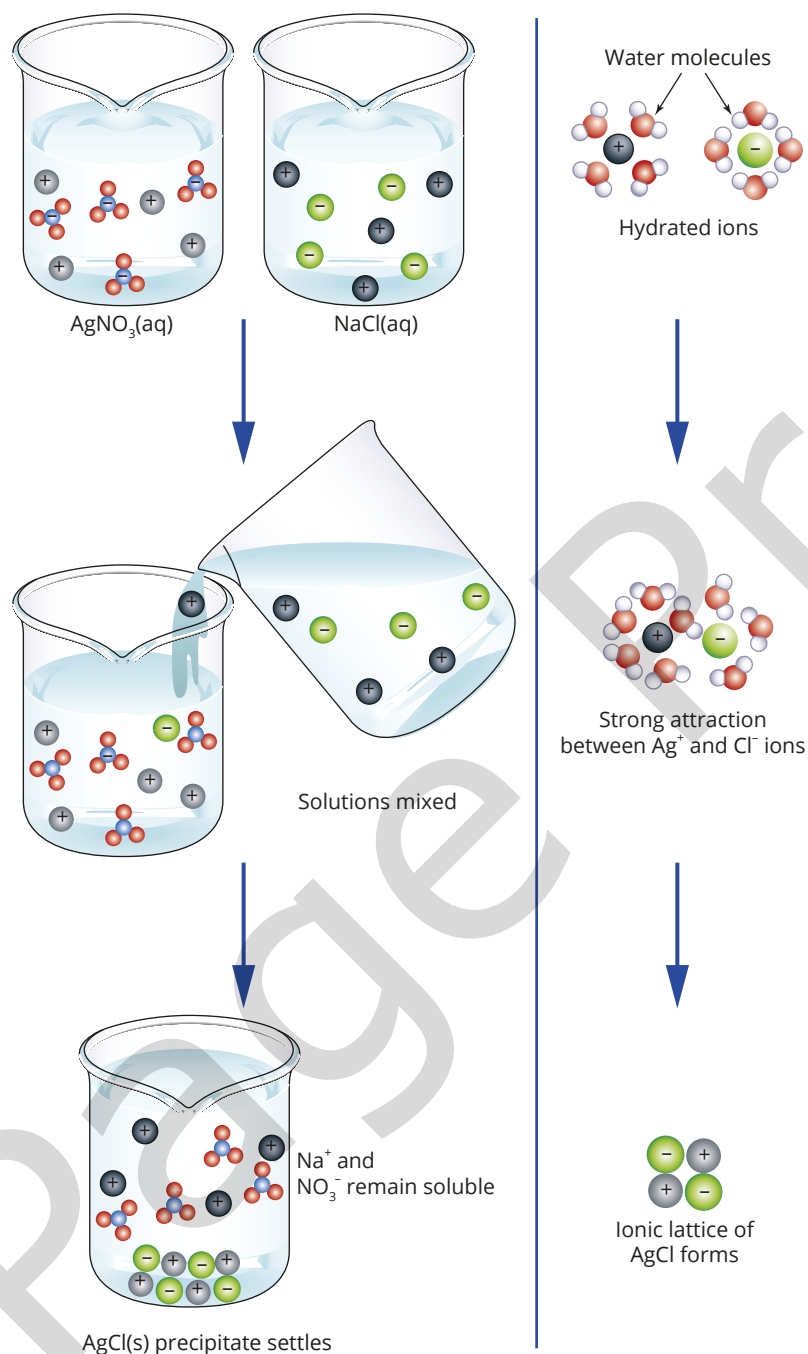


FIGURE 16.1.3 Pictorial representation of mixing aqueous solutions of sodium chloride and silver nitrate to produce a precipitate of silver chloride

There is a simple way that allows you to work out which compound will form the precipitate in a reaction between two ionic solutions.

- Write down the formula for the cation of one of the compounds, followed by its anion. Repeat the process for the second compound. For example, Ag^+ , NO_3^- , Na^+ , Cl^- as shown in column 2 of Figure 16.1.4.
- Then draw two lines. The first line joins the cation of the first solution to the anion of the second. The second line joins the anion of the first solution to the cation of the second (column two of Figure 16.1.4).
- Next, combine the ions to write the formulae for the two possible precipitates, if any (column three of Figure 16.1.4).
- Finally, use solubility tables to work out which of the two combinations of ions will result in an insoluble compound. This will be the precipitate (column four of Figure 16.1.4). The other ions will remain in solution.

	Ions in solution	Possible precipitates	Actual precipitate, if any, from solubility tables
Solution 1	Ag^+ and NO_3^-	AgCl or NaNO_3	AgCl
Solution 2	Na^+ and Cl^-		

FIGURE 16.1.4 Process for identifying the precipitate, if any, when two aqueous ionic solutions are mixed

It is important for chemists to be able to predict whether a precipitate will form in a reaction and what this precipitate will be. Worked example 16.1.1 takes you through the process of predicting the products of a precipitation reaction.

Worked example 16.1.1

PREDICTING THE PRODUCTS OF A PRECIPITATION REACTION

What precipitate, if any, will be produced when solutions of potassium hydroxide and lead(II) nitrate are added together? You will need to refer to the solubility tables (Tables 15.5.1 and 15.5.2 on page xxx or in your data booklet) to complete this question.

Thinking	Working			
Identify which ions are produced by each of the ionic compounds in the mixture.		Ions in solution	Possible precipitates	Actual precipitate, if any, from solubility tables
	Solution 1	K^+ and OH^-		
	Solution 2	Pb^{2+} and NO_3^-		
Identify which two new combinations of cations and anions are possible in the mixture of the solutions.		Ions in solution	Possible precipitates	Actual precipitate, if any, from solubility tables
	Solution 1	K^+ and OH^-	KNO_3 or $\text{Pb}(\text{OH})_2$	
	Solution 2	Pb^{2+} and NO_3^-		

Use the solubility tables to check which, if any, of these combinations will produce an insoluble compound.

	Ions in solution	Possible precipitates	Actual precipitate, if any, from solubility tables
Solution 1	K^+ and OH^-	KNO_3 or $Pb(OH)_2$	$Pb(OH)_2$
Solution 2	Pb^{2+} and NO_3^-		

Compounds containing potassium ions are usually soluble, so potassium nitrate will not form a precipitate.

Compounds containing hydroxide ions are usually insoluble, so lead(II) hydroxide will form a precipitate.

Worked example: Try yourself 16.1.1

PREDICTING THE PRODUCTS OF A PRECIPITATION REACTION

What precipitate, if any, will be produced when solutions of sodium sulfide (Na_2S) and copper(II) nitrate ($Cu(NO_3)_2$) are added together? You will need to refer to the solubility tables (Tables 15.5.1 and 15.5.2 on page xxx or in your data booklet) to complete this question.

CHEMFILE

Limescale accumulation

Have you ever wondered where that flaky white build-up on the element of your kettle comes from?

When you boil water in the kettle, ions present in the water can precipitate out, leaving a white coating called limescale on the element (Figure 16.1.5).



FIGURE 16.1.5 The accumulation of limescale on domestic kettles is the result of precipitation of calcium carbonate as water is repeatedly boiled.

The amount of build-up depends on the type of water treatment in your area. Areas that have hard water (i.e. high levels of dissolved ions) have a bigger problem with limescale.

Limescale mostly consists of calcium carbonate ($CaCO_3$) that precipitates out as a crystalline solid when the water is boiled. High levels of limescale can build up in pipes and eventually restrict or even block the flow of water (Figure 16.1.6).



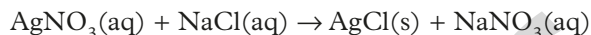
FIGURE 16.1.6 Accumulation of limescale in pipes is a result of precipitation of $CaCO_3$.

Limescale can be a big problem in the home. A coating as thin as 1.5 mm over a heating element can reduce its efficiency by as much as 12%.

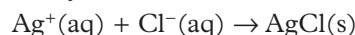
Many modern houses use ion filters or water conditioners to remove dissolved ions from the water and reduce the accumulation of limescale.

WRITING IONIC EQUATIONS FOR PRECIPITATION REACTIONS

For any reaction, including precipitation reactions, the most accurate way to represent the reaction is to show only those species (ions or molecules) that participate in the reaction. For the reaction between silver nitrate and sodium chloride solutions, a **full equation** with the complete formulae of all the substances shown in the reaction can be written. The state symbols for each of the species in the chemical reaction must be shown:

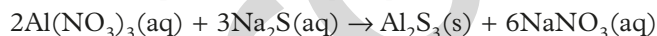


However, the essential feature of the reaction between silver nitrate and sodium chloride is the combination of silver ions and chloride ions to form a precipitate. This reaction is more accurately summarised in an **ionic equation**:



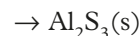
Note that spectator ions are not included in an ionic equation. Only the species that change are included. Following the process outlined in Figure 16.1.4, writing the ionic equation requires writing on the reactant side the formulae of the ions joined by the line between the ions in the precipitate and the formula for the precipitate on the product side, and then balancing the equation if needed.

Another way to write an ionic equation for a precipitation reaction is shown in the following example of the reactions between solutions of aluminium nitrate and sodium sulfide. The full equation for the precipitation reaction is:

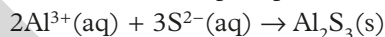


To write an ionic equation:

- 1 Write down the formula of the precipitate on the right-hand side of the page. Include a symbol of state. Place an arrow to the left of it.



- 2 To the left of this formula, add the formulae of the ions that form the precipitate, in the ratio shown by the formula of the precipitate. Include state symbols.



- 3 Check that the equation is balanced.

Worked example 16.1.2

WRITING EQUATIONS FOR PRECIPITATION REACTIONS

Write a balanced ionic equation for the reaction between iron(III) nitrate and sodium sulfide, in which the precipitate is iron(III) sulfide. Identify the spectator ions in this reaction.	
Thinking	Working
Identify the ions present in each solution.	Iron(III) nitrate solution has $\text{Fe}^{3+}(\text{aq})$ and $\text{NO}_3^-(\text{aq})$ ions. Sodium sulfide solution has $\text{Na}^+(\text{aq})$ and $\text{S}^{2-}(\text{aq})$ ions.
Write down the formula of the precipitate on the right-hand side. Include the state symbol (s). Place an arrow to the left of it.	$\rightarrow \text{Fe}_2\text{S}_3(\text{s})$
To the left of this formula, add the formulae of the ions that form the precipitate, in the ratio shown by the formula of the precipitate. Include state symbols. Check the equation is balanced.	$2\text{Fe}^{3+}(\text{aq}) + 3\text{S}^{2-}(\text{aq}) \rightarrow \text{Fe}_2\text{S}_3(\text{s})$
Write the formulae of the ions that do not form a precipitate in the reaction. These are the spectator ions.	$\text{Na}^+(\text{aq})$ and $\text{NO}_3^-(\text{aq})$ are spectator ions.

Worked example: Try yourself 16.1.2

WRITING EQUATIONS FOR PRECIPITATION REACTIONS

Write a balanced ionic equation for the reaction between copper(II) sulfate and sodium hydroxide, in which the precipitate is copper(II) hydroxide. Identify the spectator ions in this reaction.

Worked example 16.1.3

IDENTIFYING PRECIPITATES AND WRITING EQUATIONS FOR PRECIPITATION REACTIONS

Identify the precipitate, if any, when solutions of copper(II) chloride and sodium hydroxide are mixed. Write the balanced ionic equation for the formation of any precipitate. Identify the spectator ions in any reaction.

Thinking	Working
Identify the ions present in each solution.	Copper(II) chloride solution has $\text{Cu}^{2+}(\text{aq})$ and $\text{Cl}^{-}(\text{aq})$ ions. Sodium hydroxide solution has $\text{Na}^{+}(\text{aq})$ and $\text{OH}^{-}(\text{aq})$ ions.
Identify the possible products.	copper(II) hydroxide and sodium chloride
Use the solubility rules to determine if one of the products is insoluble.	Yes: copper(II) hydroxide
Write down the formula of the precipitate on the right-hand side. Include the state symbol (s). Place an arrow to the left of it.	$\rightarrow \text{Cu}(\text{OH})_2(\text{s})$
To the left of this formula, add the formulae of the ions that form the precipitate, in the ratio shown by the formula of the precipitate. Include state symbols. Check the equation is balanced.	$\text{Cu}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s})$
Write the formulae of the ions that do not form a precipitate in the reaction. These are the spectator ions.	$\text{Na}^{+}(\text{aq})$ and $\text{Cl}^{-}(\text{aq})$ are spectator ions.

Worked example: Try yourself 16.1.3

IDENTIFYING PRECIPITATES AND WRITING EQUATIONS FOR PRECIPITATION REACTIONS

Identify the precipitate, if any, when solutions of potassium carbonate and copper(II) sulfate are mixed. Write the balanced ionic equation for the formation of any precipitate. Identify the spectator ions in any reaction.

CHEMISTRY IN ACTION

The chemistry of colour

If you have ever walked through an art supplies store, you will have noticed displays selling paints with names that sound as though they belong in a chemistry laboratory. Names such as lead yellow, titanium white and cobalt blue are just a few of the colours still in use today (Figure 16.1.7).

The names are not just for show. The colours that they represent have their basis in chemistry. All paints consist of a pigment (the colour) and a binder that holds the pigments in a suspension. Historically, paints were purchased as powdered pigments like those in Figure 16.1.8.

Artists would mix these pigments with their own binders, such as linseed oil, to produce the required paints. Ancient civilisations such as the Egyptians would grind minerals like lapis lazuli and ochre from the earth to make pigments. Early alchemists were able to manufacture synthetic pigments using precipitation reactions, collecting the coloured precipitates through filtration.

Prussian blue was a very popular colour through the 19th century and is made from a precipitate of iron(III) hexacyanoferrate(II). It has been used extensively in ceramics and painting, including the famous Japanese block print from the 1830s *The Great Wave off Kanagawa* (Figure 16.1.9).



FIGURE 16.1.9 *The Great Wave off Kanagawa* (c. 1830–1833). This woodblock print is by the Japanese artist Hokusai (1760–1849). The deep blue tones were painted with a paint known as Prussian blue made from iron(III) hexacyanoferrate(II).

Pigments based on ionic precipitates provided a much wider array of colours to artists. This paved the way for the more flamboyant use of colour seen in the work of the French impressionists. Cadmium sulfide was used to make red pigments and cadmium red is still used widely today. Viridian, a deep green pigment, is manufactured from chromium(III) oxide dihydrate.

So the next time you visit an art gallery, take a moment to think about the complex chemistry that went into bringing those colours to life.



FIGURE 16.1.7 Paint colours are often named after the pigments that were once used to make them. Many of these pigments were the product of precipitation reactions.



FIGURE 16.1.8 An array of different coloured pigments. Historically, pigments for paints were collected from minerals and other sources. However, in the 19th century a better knowledge of chemistry was used to manufacture pigments more reliably and in a larger variety of colours.

16.1 Review

SUMMARY

- A precipitation reaction occurs when two solutions of compounds are mixed and a solid product is formed. The solid product is called a precipitate.
- Solubility tables can be used to predict which compound, if any, will precipitate in a reaction.
- Ions that are not directly involved in the formation of the precipitate are called spectator ions.
- Full and ionic equations can be written for precipitation reactions.
- Precipitation reactions are most accurately represented by ionic equations.
- Ionic equations do not include spectator ions.

KEY QUESTIONS

- 1 Use the solubility tables to identify the precipitate formed, if any, when the following solutions are mixed.
 - a silver nitrate and potassium carbonate
 - b potassium hydroxide and lead(II) nitrate
 - c magnesium chloride and sodium sulfide
 - d sodium chloride and iron(II) sulfate
- 2
 - a Name the precipitate formed when aqueous solutions of the following compounds are mixed.
 - i K_2S and MgCl_2
 - ii CuCl_2 and AgNO_3
 - iii KOH and AlCl_3
 - iv MgSO_4 and NaOH
 - b Write a balanced ionic equation for each reaction.
- 3 Write an ionic equation for any precipitate formed when the following aqueous solutions are mixed.
 - a $\text{AgNO}_3(\text{aq}) + \text{NaI}(\text{aq}) \rightarrow$
 - b $\text{CuSO}_4(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow$
 - c $(\text{NH}_4)_2\text{SO}_4(\text{aq}) + \text{BaCl}_2(\text{aq}) \rightarrow$
 - d $\text{K}_2\text{SO}_4(\text{aq}) + \text{Pb}(\text{NO}_3)_2(\text{aq}) \rightarrow$
 - e $\text{CaCl}_2(\text{aq}) + \text{Na}_3\text{PO}_4(\text{aq}) \rightarrow$
 - f $\text{NaOH}(\text{aq}) + \text{Pb}(\text{NO}_3)_2(\text{aq}) \rightarrow$
- 4 Where a precipitate formed in Question 3, identify the spectator ions.

16.2 Concentration of solutions

The **concentration** of a solution describes the relative amount of solute and solvent present in the solution. A solution in which the ratio of solute to solvent is high is said to be concentrated. Cordial that has not had any water added to it is an example of a **concentrated solution**. A solution in which the ratio of solute to solvent is low is said to be a **dilute solution**. A quarter of a teaspoon of sugar dissolved in a litre of water will produce a dilute solution that tastes slightly sweet.

‘Concentrated’ and ‘dilute’ are general terms. However, sometimes you need to know the actual concentration of a solution—the exact ratio of solute to solvent. The use of exact solution concentrations is important in the prescription of medicines, chemical manufacturing and chemical analysis.

UNITS OF CONCENTRATION

Chemists use different measures of concentration depending on the situation. The most common measures describe the amount of solute in a given amount of solution.

Some examples of units of concentration are:

- mass of solute per litre of solution (grams per litre, g L^{-1})
- moles of solute per litre of solution (moles per litre, mol L^{-1}). (You will look more closely at this unit of concentration in Section 16.3.)
- parts per million (ppm)
- percentage by mass.

Units of concentration measured in this way have two parts.

- The first part gives information about how much solute there is.
- The second part gives information about how much solution there is.

For example, if a solution contains sodium chloride (NaCl) with a concentration of 17 g L^{-1} , then (first part) 17 g of NaCl is dissolved in every (second part) 1 L of the solution.

You will now look at how to perform calculations of concentration using different units.

Concentration in grams per litre (g L^{-1})

The concentration of a solution in grams per litre (g L^{-1}) indicates the mass, in grams, of solute dissolved in 1 litre of the solution. For example, if the concentration of sodium chloride in seawater is 20 g L^{-1} , this means that in 1 L of seawater there is 20 g of sodium chloride. A formula used to calculate the concentration in g L^{-1} is:

$$\text{concentration (g L}^{-1}\text{)} = \frac{\text{mass of solute (in g)}}{\text{volume of solution (in L)}}$$

Worked example 16.2.1

CALCULATING CONCENTRATION IN g L^{-1} (GRAMS PER LITRE)

What is the concentration, in g L^{-1} , of a solution containing 8.00 g of sodium chloride in 500 mL of solution?	
Thinking	Working
Change the volume of solution so it is expressed in litres.	$500 \text{ mL} = \frac{500}{1000}$ $= 0.500 \text{ L}$
Calculate the concentration in g L^{-1} .	$c = \frac{\text{mass of solute (in g)}}{\text{volume of solution (in L)}}$ $= \frac{8.00}{0.500}$ $= 16.0 \text{ g L}^{-1}$

Worked example: Try yourself 16.2.1

CALCULATING CONCENTRATION IN g L^{-1} (GRAMS PER LITRE)

What is the concentration, in g L^{-1} , of a solution containing 5.00 g of glucose in 250 mL of solution?

Concentration in parts per million (ppm)

When very small quantities of solute are dissolved to form a solution, the concentration is often measured in **parts per million (ppm)**. For example, the concentration of mercury in fish that is safe for consumption is usually expressed in parts per million. The maximum concentration allowed for sale in Australia is 1 ppm (Figure 16.2.1).

In simple terms, the concentration in parts per million can be thought of as the mass in grams of solute dissolved in 1 000 000 g of solution. This can also be expressed as the mass in milligrams of solute dissolved in 1 kg of solution, because there are 1 million milligrams in a kilogram.

For example, a solution of sodium chloride that has a concentration of 154 ppm contains 154 mg of sodium chloride dissolved in 1 kg of solution. A formula used to calculate the concentration of a solution in ppm is:

$$\text{concentration (ppm)} = \frac{\text{mass of solute (in mg)}}{\text{mass of solution (in kg)}}$$



FIGURE 16.2.1 Fish sold in Australia must have no more than 1 ppm of mercury (Australia New Zealand Food Standards Code).

Worked example 16.2.2

CALCULATING CONCENTRATION IN PARTS PER MILLION (PPM)

A saturated solution of calcium carbonate was found to contain 0.0198 g of calcium carbonate dissolved in 2000 g of solution. Calculate the concentration, in ppm, of calcium carbonate in the saturated solution.

Thinking	Working
Calculate the mass of solute in mg. Remember: $\text{mass (in mg)} = \text{mass (in g)} \times 1000$	Mass of solute (calcium carbonate) in mg: $= 0.0198 \times 1000$ $= 19.8 \text{ mg}$
Calculate the mass of solution in kg. Remember: $\text{mass (in kg)} = \frac{\text{mass (in g)}}{1000}$	Mass of solution in kg $= \frac{2000}{1000}$ $= 2.000 \text{ kg}$
Calculate the concentration of the solution in mg kg^{-1} . This is the same as concentration in ppm.	Concentration of calcium carbonate in ppm $= \frac{\text{mass of solute (in mg)}}{\text{mass of solution (in kg)}}$ $= \frac{19.8}{2.000}$ $= 9.90 \text{ mg kg}^{-1}$ $= 9.90 \text{ ppm}$

Worked example: Try yourself 16.2.2

CALCULATING CONCENTRATION IN PARTS PER MILLION (PPM)

A sample of tap water was found to contain 0.0537 g of NaCl per 250.0 g of solution. Calculate the concentration of NaCl in parts per million (ppm).

EXTENSION

Other units of concentration

You might have noticed symbols such as w/w, v/v or w/v on the labels of some foods, drinks and pharmaceuticals. These symbols represent other concentration units based on masses and volumes of solutes and solutions. These are useful in practical situations because people are familiar with these quantities (Figure 16.2.2).



FIGURE 16.2.2 Consumer products from hardware stores and supermarkets show a wide range of concentration units on their labels.

Percentage by mass (w/w)

Percentage by mass describes the mass of solute, measured in grams, present in 100g of the solution.

Normal saline solution for washing contact lenses has a concentration of 0.9%(w/w). This can also be written as 0.9%(m/m). The abbreviation w/w indicates that the percentage is based on the weights or, more correctly, masses of both solute and solution. A concentration of 0.9% indicates that there is 0.9 g of sodium chloride dissolved in 100 g of solution.

Percentage by volume (v/v)

The abbreviation v/v indicates that the percentage is based on volumes of both solute and solution. The same units must be used to record both volumes. Percentage by volume is a more convenient unit to use than w/w when the solute is a liquid.

Just like percentage by mass, percentage by volume is frequently expressed as volume per 100mL of solution. For example, the wine label in Figure 16.2.3 shows 13.5% alc./vol. This means the wine contains 13.5% alcohol (ethanol) by volume (13.5%(v/v)). There will be 13.5mL of alcohol in 100mL of the wine.

For example, a 200mL glass of champagne contains 28mL of alcohol. The concentration as %(v/v) of alcohol in this solution can be calculated as follows:

$$\begin{aligned}\text{concentration (\% v/v)} &= \frac{\text{volume of solute (in mL)}}{\text{volume of solution (in mL)}} \times 100\% \\ &= \frac{28}{200} \times 100\% \\ &= 14\%(\text{v/v})\end{aligned}$$



FIGURE 16.2.3 The alcohol content of this wine is 13.5% v/v.

Percentage mass/volume (w/v)

Percentage mass/volume describes the mass of solute, measured in grams, present in 100 mL of the solution.

For example, if a solution of plant food contains a particular potassium compound at a concentration of 3%(w/v), this indicates that there is 3g of potassium in 100mL of solution.

CHEMFILE

Salinity in Western Australia

The south-west region of Western Australia, like other parts of Australia, has increasing salt levels in both surface and ground water. Salt has accumulated in south-west soils over hundreds of thousands of years due to wind and rain carrying it from the ocean. Prior to European settlement the salt was mostly undissolved in the soil but with clearing of trees which used much of the rainfall, more rain now enters the ground water causing a rise in the water table which in turn dissolves salt in the soil bringing it closer to the surface. As ground water rises, some enters the surface water bringing its salt with it. For example, in the 1940s the Warren River had about 300 mg L^{-1} of salt but now the level is around $800\text{--}900 \text{ mg L}^{-1}$, and the Blackwood River has gone from about 500 mg L^{-1} to 2000 mg L^{-1} over the past 50 years.

Water is classified as fresh if the salt concentration is less than 500 mg L^{-1} , marginal if it is between $500\text{--}1000 \text{ mg L}^{-1}$ and saline above 2000 mg L^{-1} . Marginal water can be used for most irrigation purposes but the ecosystem begins to suffer, and saline water may be suitable for some livestock but it is no longer suitable for irrigation or cropping.

CHEMFILE

Saline drip

A 'saline drip' is sometimes used during medical procedures to replenish body fluids in patients (Figure 16.2.4). The solution used contains sodium chloride, commonly with a concentration of $0.9\%(\text{m/v})$. That means that 100 mL of the solution contains 0.9 g of sodium chloride.



FIGURE 16.2.4 A saline drip is a solution of sodium chloride.

16.2 Review

SUMMARY

- The concentration of a solution is defined as the quantity of solute dissolved in a quantity of solution.
- Concentration units have two parts. The first part provides information about the quantity of solute. The second part provides information about the quantity of solution.
- The concentrations of solutions can be expressed in different units including g L^{-1} , mol L^{-1} and ppm.

KEY QUESTIONS

- 1 What is the concentration, in g L^{-1} , of a 60 mL solution that contains 5.0 g of sugar?
A 300
B 0.83
C 12
D 83
- 2 Calculate the concentration of the following solutions in ppm.
 - a 25 mg of CaCl_2 dissolved in 5.0 kg of solution
 - b 1.25 g of lead nitrate dissolved in 2000 g of solution
 - c $4.0 \times 10^{-3} \text{ g}$ of MgSO_4 dissolved in 150 g of solution
- 3 The nutrition panel on a flavoured milk drink states that it contains 35.0 g of sugar and 7.5 g of fat in every 250 mL serving. What are the concentrations of sugar and fat expressed as $\%(\text{w/v})$?
- 4 Fish sold in Australia can have a maximum concentration of mercury of 1 ppm . A fish catch is sampled and a fish of mass 1.26 kg is found to contain 1.092 mg of Hg. Can the fish catch be sold in Australia?

16.3 Molar concentration

In the previous section you saw that units of concentration such as g L^{-1} , $\%(\text{w/w})$ and $\%(\text{v/v})$ are commonly used on the labels of consumer products found in supermarkets, pharmacies and hardware stores. Another unit of concentration that is commonly used by chemists is mol L^{-1} .

In this section, you will learn how to carry out calculations using this unit of concentration.

CONCENTRATION IN MOLES PER LITRE

Concentrations expressed in moles per litre (mol L^{-1}) allow chemists to compare the amount, in moles, of atoms, molecules or ions present in a given volume of solution.

For example, a 0.1 mol L^{-1} solution of hydrochloric acid would contain 0.1 moles of HCl in 1 L of the solution. (A bottle of HCl of this concentration is shown in Figure 16.3.1.)

The concentration, in mol L^{-1} , of a solution can be calculated as follows:

$$\text{concentration (mol L}^{-1}\text{)} = \frac{\text{amount of solute (in mol)}}{\text{volume of solution (in L)}}$$

$$c = \frac{n}{V}$$

where c is the concentration (mol L^{-1}), n is the amount (mol) and V is the volume (L).

The concentration of a solution in moles per litre is often referred to as the **molarity**, or molar concentration of the solution. Molarity has the unit mol L^{-1} , which is sometimes represented as M.

A solution containing 1 mole of solute dissolved in 1 litre of solution can therefore be described in several different ways. You can say that the solution:

- has a concentration of 1 mole per litre
- has a concentration of 1 mol L^{-1}
- is 1 molar
- is 1 M
- has a molarity of 1 M.

Worked example 16.3.1

CALCULATING MOLAR CONCENTRATIONS (MOLARITY)

Calculate the molar concentration of a solution that contains 0.105 mol of potassium nitrate dissolved in 200 mL of solution.

Thinking

Convert the given volume to litres.

Working

$$V(\text{KNO}_3) = \frac{200}{1000}$$

$$= 0.200 \text{ L}$$

Calculate the molar concentration using the formula:

$$c = \frac{n}{V}$$

$$c(\text{KNO}_3) = \frac{n}{V}$$

$$= \frac{0.105}{0.200}$$

$$= 0.525 \text{ mol L}^{-1}$$

Worked example: Try yourself 16.3.1

CALCULATING MOLAR CONCENTRATIONS (MOLARITY)

Calculate the molar concentration of a solution that contains 0.24 mol of glucose dissolved in 500 mL of solution.



FIGURE 16.3.1 These containers contain solutions with concentrations shown in units of mol L^{-1} . The mol L^{-1} unit is often written as M.

i The symbol M was used earlier to represent molar mass. Take care not to confuse the two different uses, one is a quantity symbol, the other a unit symbol.

i Molarity is calculated using the formula $c = \frac{n}{V}$, where c is the concentration in mol L^{-1} , n is the amount in moles and V is the volume in litres.

Calculating molarity given the mass of solute

Sometimes you will know the mass of a solute and volume of solution and want to calculate the molarity. Two main calculations are involved.

- 1 Calculate the number of moles of solute from its mass.
- 2 Calculate the concentration using the number of moles and the volume (in litres).

Worked example 16.3.2

CALCULATING MOLARITY GIVEN THE MASS OF SOLUTE

Calculate the concentration, in mol L^{-1} , of a solution that contains 16.8 mg of silver nitrate (AgNO_3) dissolved in 150 mL of solution.	
Thinking	Working
Convert the volume to litres.	$V(\text{AgNO}_3) = \frac{150}{1000}$ $= 0.150 \text{ L}$
Convert the mass to grams.	$m(\text{AgNO}_3) = \frac{16.8}{1000}$ $= 0.0168 \text{ g}$
Calculate the molar mass of the solute. To do this, add up the atomic masses of all the atoms in the compound.	$M(\text{AgNO}_3) = 107.9 + 14.0 + (3 \times 16.0)$ $= 169.9 \text{ g mol}^{-1}$
Calculate the number of mol of solute using the formula: $n = \frac{m}{M}$	$n(\text{AgNO}_3) = \frac{m}{M}$ $= \frac{0.0168}{169.9}$ $= 9.89 \times 10^{-5} \text{ mol}$
Calculate the molar concentration using the formula: $c = \frac{n}{V}$	$c(\text{AgNO}_3) = \frac{n}{V}$ $= \frac{9.89 \times 10^{-5}}{0.150}$ $= 6.59 \times 10^{-4} \text{ mol L}^{-1}$

Worked example: Try yourself 16.3.2

CALCULATING MOLARITY GIVEN THE MASS OF SOLUTE

Calculate the concentration, in mol L^{-1} , of a solution that contains 4000 mg of ethanoic (acetic) acid (CH_3COOH) dissolved in 100 mL of solution.

The formula used to calculate the molarity of a solution is $c = \frac{n}{V}$. If the formula is rearranged, it can be used to calculate the number of moles of solute in a solution of given concentration and volume:

$$n = c \times V$$

where n is the amount (mol), c is the concentration (mol L^{-1}) and V is the volume (L).

CHEMFILE

Molarity—the chemist's unit of concentration

If you ever look at the labels on household and commercial products you are very unlikely to see the concentrations of the active ingredients given in units of molarity (mol L^{-1}). The units typically used will include g/L (grams per litre), mg/g (milligrams per gram), % (w/v) (weight/volume percent) or % (w/w) (weight/weight percent). Chemists however most commonly use molarity – the number of moles in one litre of solution.

When working with solutions, chemists are often dealing with the reaction of a substance in the solution with another substance (which may also be in solution). Substances react at the molecular or particle level so chemists are more likely to be interested in the number of particles present so that the quantity of the substances in the reaction can be easily related to each other. It is simplest to relate the quantity of substances in a chemical reaction through their number of moles, and molarity allows this more readily for solutions than do other units of concentration.

Worked example 16.3.3

CALCULATING THE NUMBER OF MOLES OF SOLUTE IN A SOLUTION

Calculate the amount, in moles, of ammonia (NH_3) in 25.0 mL of a $0.3277 \text{ mol L}^{-1}$ ammonia solution.

Thinking	Working
Convert the given volume to litres.	$V(\text{NH}_3) = \frac{25.0}{1000}$ $= 0.0250 \text{ L}$
Calculate the amount of compound, in moles, using the formula: $n = c \times V$	$n(\text{NH}_3) = c \times V$ $= 0.3277 \times 0.0250$ $= 8.19 \times 10^{-3} \text{ mol}$

Worked example: Try yourself 16.6.3

CALCULATING THE NUMBER OF MOLES OF SOLUTE IN A SOLUTION

Calculate the amount, in moles, of potassium permanganate (KMnO_4), in 100 mL of a $0.0250 \text{ mol L}^{-1}$ solution of the compound.

16.3 Review

SUMMARY

- Molarity is defined as the number of moles of solute per litre of solution.
- Molarity is calculated using the formula $c = \frac{n}{V}$, where c is the concentration in mol L^{-1} , n is the amount in moles and V is the volume in litres.
- The formula $n = c \times V$ can be used to calculate the number of moles of solute in a given volume of solution.

KEY QUESTIONS

- Hydrogen peroxide solutions for hair bleaching are sold as 120 mL solutions containing 5.00 g of H_2O_2 dissolved in water. What is the molar concentration of hydrogen peroxide in this bleaching product?
A 0.81 % (w/v)
B 1.23 mol L^{-1}
C 41.7 g L^{-1}
D $1.23 \times 10^{-3} \text{ mol L}^{-1}$
- Calculate the molar concentration of the following solutions.
 - a 25.0 mL solution that contains $2.0 \times 10^{-3} \text{ mol}$ of NaCl
 - a 4.1 L solution that contains 1.23 mol of CH_3COOH
 - a $9.3 \times 10^3 \text{ L}$ solution that contains $1.8 \times 10^3 \text{ mol}$ of KCl
- Calculate the molar concentration of the following solutions.
 - an 8.0 L solution that contains 2.0 mol of NaCl
 - a 500 mL solution that contains 0.25 mol of MgCl_2
 - a 200 mL solution that contains 0.0876 mol of sucrose
- Calculate the molar concentration of a 250 mL solution that contains 5.09 g of AgNO_3 .
- Calculate the molar concentration of a 1.55 L solution that contains 1.223 g of CaCl_2 .
- Calculate the amount, in mol, of solute in each of the following solutions.
 - 0.10 L of 0.22 mol L^{-1} KOH solution
 - 10 mL of 0.64 mol L^{-1} NaI solution
 - 15.6 mL of $0.0150 \text{ mol L}^{-1}$ CuSO_4 solution
 - $1.5 \times 10^{-1} \text{ mL}$ of 5.2 mol L^{-1} HCl solution

16.4 Dilution

Many commercially available domestic and industrial products come in the form of concentrated solutions. Examples are pesticides (Figure 16.4.1), fertilisers, detergents, fruit juices, acids and other chemicals. A major reason for using concentrates is to save on transportation costs. Diluted solutions contain a lot of water and that extra mass has to be transported, which increases costs. It is also more convenient to buy concentrated products and dilute with water at home or in the workplace.

Everyday examples of dilution are:

- adding water to cordial
- a laboratory technician making a 1 mol L^{-1} solution of hydrochloric acid from a bottle of concentrated hydrochloric acid
- a home gardener diluting fertiliser concentrate to spray on the lawn
- a farmer diluting weedkiller concentrate to spray on a wheat crop
- an assistant in a commercial kitchen diluting concentrated detergent solution before using it to wash dishes.

In this section, you will learn how to calculate the concentrations of solutions before and after they have been diluted. As well, you will learn how to calculate the volume of water that needs to be added to a concentrated solution to give a diluted solution of a required concentration, and the volume of concentrated solution needed to produce a diluted solution of known volume and concentration.

CALCULATING CONCENTRATIONS WHEN SOLUTIONS ARE DILUTED

The process of adding more solvent to a solution is known as **dilution**. When a solution is diluted, its concentration is decreased.

For example, if 50 mL of water is added to 50 mL of 0.10 mol L^{-1} sugar solution, the amount of sugar remains unchanged but the volume of the solution in which it is dissolved doubles. As Figure 16.4.2 shows, this means the sugar molecules are spread further apart during the dilution process, and so the concentration of the sugar solution is decreased (it will become 0.050 mol L^{-1} in this instance).

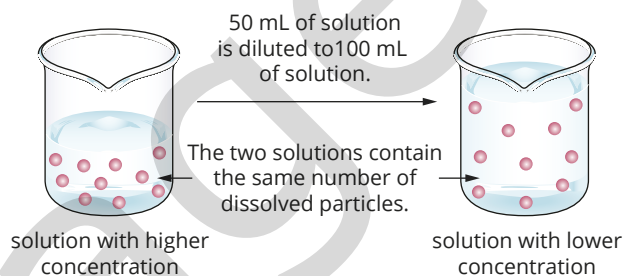


FIGURE 16.4.2 Dilution does not change the number of solute particles but the concentration of the solute decreases.

It is important to recognise that diluting a solution (by adding more solvent) does not change the amount of solute present.

Suppose you had V_1 litres of a solution and the concentration was $c_1 \text{ mol L}^{-1}$. The amount of solute, in mole, is given by:

$$n_1 = c_1 \times V_1$$

Suppose water was added to make a new volume, V_2 , and change the concentration to c_2 . The amount of solute, n_2 , in this diluted solution is given by:

$$n_2 = c_2 \times V_2$$

Since the number of moles of solute has not changed, $n_1 = n_2$; therefore:

$$c_1 V_1 = c_2 V_2$$



FIGURE 16.4.1 Pesticide solutions used in aerial spraying are prepared by diluting a concentrated solution of the pesticide compound.

i Numerical problems involving dilution can be solved using the formula: $c_1V_1 = c_2V_2$

This formula is useful when solving problems involving diluted solutions. (Note that this formula can also be used with different concentration and volume units, as long as they are the same on both sides.)

Worked example 16.4.1

QUESTIONS INVOLVING DILUTION

Calculate the concentration of the solution formed when 10.0 mL of water is added to 5.00 mL of 1.2 mol L ⁻¹ HCl.	
Thinking	Working
Write down the value of c_1 and V_1 . Note: c_1 and V_1 refer to the original solution, before water was added.	$c_1 = 1.2 \text{ mol L}^{-1}$ $V_1 = 5.00 \text{ mL}$
Write down the value of V_2 . Note: V_2 is the total volume of the original solution plus the added water.	$V_2 = 10.0 + 5.00$ $= 15.0 \text{ mL}$
Transpose the equation $c_1V_1 = c_2V_2$ to allow calculation of the concentration, c_2 , of the new solution.	$c_1V_1 = c_2V_2$ $c_2 = \frac{c_1V_1}{V_2}$
Calculate the concentration of the diluted solution.	$c_2 = \frac{1.2 \times 5.00}{15.0}$ $= 0.40 \text{ mol L}^{-1}$

Worked example: Try yourself 16.4.1

QUESTIONS INVOLVING DILUTION

Calculate the concentration of the solution formed when 95.0 mL of water is added to 5.00 mL of 0.500 mol L⁻¹ HCl.

CALCULATING VOLUMES WHEN SOLUTIONS ARE DILUTED

When a laboratory technician is preparing a diluted solution from a concentrated solution to use in a laboratory procedure, it is usual that they know the concentration and volume of the diluted solution they want to prepare. As well, they will know the concentration of the concentrated solution that is being used to make the dilute solution. Thus to make their diluted solution they need to calculate how much water to add to the concentrated solution.

For example, sulfuric acid is typically provided to laboratories as a 98% solution which has a molar concentration of 18.4 mol L⁻¹. The concentrations at which the acid is more typically used in laboratory work are around the 1 mol L⁻¹ range.

Worked example 16.4.2

QUESTIONS INVOLVING DILUTION

Calculate the volume of 18.4 mol L ⁻¹ sulfuric acid solution required to prepare 5.00 L of a 1.50 mol L ⁻¹ solution. What volume of water must be added to the 18.4 mol L ⁻¹ sulfuric acid solution to give the required volume of the final solution?	
Thinking	Working
Write down the value of c_1 and V_1 . Note: c_1 and V_1 refer to the final solution, after water was added.	$c_1 = 1.50 \text{ mol L}^{-1}$ $V_1 = 5.00 \text{ L}$

Write down the value of c_2 . Note: c_2 is the concentration of the original solution.	$c_2 = 18.4 \text{ mol L}^{-1}$
Transpose the equation $c_1V_1 = c_2V_2$ to allow calculation of the concentration, V_2 , of the new solution.	$c_1V_1 = c_2V_2$ $V_2 = \frac{c_1V_1}{c_2}$
Calculate the volume of the concentrated solution.	$V_2 = \frac{1.50 \times 5.00}{18.4}$ $= 0.408 \text{ L}$
Calculate the volume of water required by finding the difference between the final volume of the diluted solution and the volume of concentration solution needed.	$V(\text{water}) = 5.00 - 0.408$ $= 4.59 \text{ L}$

Worked example: Try yourself 16.4.2

QUESTIONS INVOLVING DILUTION

Calculate the volume of 11.5 mol L^{-1} hydrochloric acid solution required to prepare 2.5 L of a 0.50 mol L^{-1} solution. What volume of water must be added to the 11.5 mol L^{-1} hydrochloric acid solution to give the required volume of the final solution?

CHANGING THE UNITS OF CONCENTRATION

At times it is useful to change, or 'convert', concentration values from one unit to another unit. One way to do this is to assume that you have one litre of the solution. Worked example 16.4.3 shows how this can be done.

Worked example 16.4.3

CONCENTRATION UNIT CONVERSIONS

What is the concentration, in ppm, of a $0.00200 \text{ mol L}^{-1}$ solution of NaCl? Note that concentration in ppm is the same as mg L^{-1} . If the density of the solution is 1 kg L^{-1} , which can be assumed for solutions of low concentrations.	
Thinking	Working
Calculate the number of moles of solute in 1.00 L of the solution.	$n(\text{NaCl}) = c \times V$ $= 0.00200 \times 1.00$ $= 0.00200 \text{ mol}$
Calculate the mass, in grams, of solute in 1.00 L of the solution.	$M(\text{NaCl}) = 35.5 + 23.0$ $= 58.5 \text{ g mol}^{-1}$ $m(\text{NaCl}) = n \times M$ $= 0.00200 \times 58.5$ $= 0.117 \text{ g}$
Calculate the mass, in mg, of solute in 1.00 L of the solution.	$m(\text{NaCl}) = 0.117 \times 1000$ $= 117 \text{ mg}$
Express the concentration of the solute in ppm.	$c(\text{NaCl}) = 117 \text{ ppm}$

Worked example: Try yourself 16.4.3

CONCENTRATION UNIT CONVERSIONS

What is the concentration, in ppm, of a $0.0100 \text{ mol L}^{-1}$ solution of NaOH? Note that concentration in ppm is the same as mg L^{-1} . If the density of the solution is 1 kg L^{-1} , which can be assumed for solutions of low concentrations.

16.4 Review

SUMMARY

- Numerical problems involving dilution can be solved using the formula:
$$c_1V_1 = c_2V_2$$
- where c is concentration and V is volume.
- When using this formula, c_1 and c_2 must be in the same units of concentration and V_1 and V_2 must be in the same units of volume.
- One type of concentration unit can be converted to another unit in simple steps. For example, units of $\%(w/w)$ can be converted to units of mol L^{-1} .

KEY QUESTIONS

- Calculate the final concentration of each of the following diluted solutions.
 - 10.0 mL of water is added to 5.0 mL of 1.2 mol L^{-1} HCl
 - 1.0 L of water is added to 3.0 L of 0.10 mol L^{-1} HCl
 - 5.0 mL of 0.50 mol L^{-1} HCl is added to 95.0 mL of water
- What volume of 10 mol L^{-1} hydrochloric acid would be required to prepare 250 mL of a 0.30 mol L^{-1} HCl solution?
 - 7.5 L
 - 133 L
 - 0.133 L
 - 0.0075 L
- The concentration of a solution of ammonia (NH_3) is $1.5\%(w/v)$. What is the molar concentration of a solution produced by diluting 25.0 mL of this solution with 250 mL of water?
- The concentration of a solution of NaOH is $17.0\%(w/v)$. What is the concentration of this solution in mol L^{-1} ?

16.5 Calculations involving reactions in solutions

As you learnt earlier in this chapter, water is an exceptional solvent for a large variety of substances. In this section, you will learn to calculate amounts of the reactants or products involved in chemical reactions that occur in solution. These reactions can involve precipitation reactions as well as other types of reactions. Using your knowledge of the mole concept, you will be able to calculate the mass or concentration of products formed or the amounts of reactants consumed (Figure 16.5.1).



FIGURE 16.5.1 The reaction of a solution of mercury(II) ethanoate with a solution of sodium iodide. When colourless aqueous solutions of each reactant are mixed, they produce a red precipitate, mercury(II) iodide.

CALCULATING THE MASS OF A PRECIPITATE IN A REACTION

Several steps are involved in calculating the mass of a precipitate produced in a precipitation reaction:

- 1 Write a balanced equation for the reaction.
- 2 Calculate the number of moles of any reactant that has reacted completely, using the information supplied in the question, and the formula $n = c \times V$ or $n = \frac{m}{M}$, depending on the information.
- 3 Using the mole ratios in the equation, calculate the number of moles of the precipitate.
- 4 Calculate the mass of the precipitate using $m = n \times M$.

Figure 16.5.2 summarises this process. The worked example below will help you to understand these steps.

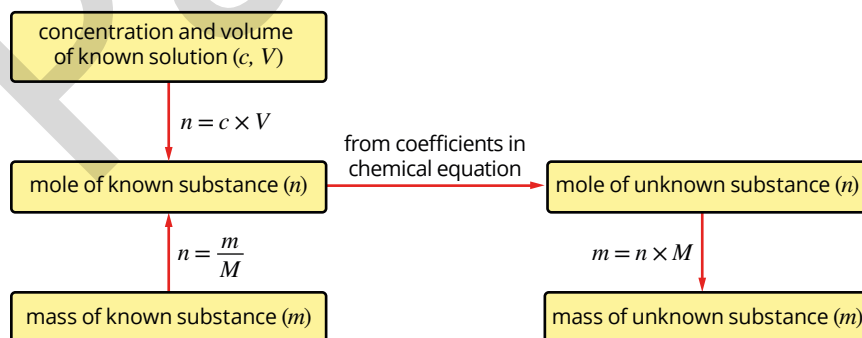


FIGURE 16.5.2 A flow chart for mass–mass or solution volume–mass stoichiometric calculations is helpful when trying to solve these problems.

Worked example 16.5.1

STOICHIOMETRY PROBLEMS INVOLVING SOLUTIONS

Calculate the mass of silver chloride precipitated when 25.0 mL of a 0.450 mol L ⁻¹ solution of calcium chloride reacts completely with a solution of silver nitrate.	
Thinking	Working
Write a balanced equation for the reaction.	$\text{CaCl}_2(\text{aq}) + 2\text{AgNO}_3(\text{aq}) \rightarrow 2\text{AgCl}(\text{s}) + \text{Ca}(\text{NO}_3)_2(\text{aq})$
Calculate the number of moles of the known substance using $n = c \times V$. The 'known substance' is the one you are provided information about in the question.	$V(\text{CaCl}_2) = 0.0250 \text{ L}$ $n(\text{CaCl}_2) = c \times V$ $= 0.450 \times 0.0250$ $= 0.0113 \text{ mol}$
Write the mole ratio for the known and unknown substances.	$\text{mole ratio} = \frac{n(\text{AgCl})}{n(\text{CaCl}_2)} = \frac{2}{1}$
Calculate the number of moles of the unknown substance using: $n(\text{unknown}) = n(\text{known}) \times \text{mole ratio}$ The 'unknown substance' is the one whose mass you are required to calculate.	$n(\text{AgCl}) = 0.0113 \times \frac{2}{1}$ $= 0.0225 \text{ mol}$
Calculate the mass of the unknown substance using: $m(\text{unknown}) = n(\text{unknown}) \times \text{molar mass}$	$M(\text{AgCl}) = 143.4 \text{ g mol}^{-1}$ $m(\text{AgCl}) = 0.0225 \times 143.4$ $= 3.23 \text{ g}$

Worked example: Try yourself 16.5.1

STOICHIOMETRY PROBLEMS INVOLVING SOLUTIONS

Calculate the mass of aluminium hydroxide precipitated when 45.0 mL of a 0.200 mol L⁻¹ solution of sodium hydroxide reacts completely with a solution of aluminium chloride.

CALCULATING OTHER QUANTITIES IN A REACTION

Not only can the mass of the precipitate formed from mixing two aqueous solutions be calculated but the quantities of other products formed or of reactants consumed can also be found.

- 1 Calculate the number of moles of the known substance, either a reactant that has reacted completely, or the product, using the formula $n = c \times V$ or $n = \frac{m}{M}$, depending on the information supplied in the question.
- 2 Using the mole ratios in the equation, calculate the number of moles of the unknown.
- 3 Calculate the quantity of the unknown by rearranging $n = c \times V$ or $n = \frac{m}{M}$ as needed.

Figure 16.5.3, an extension of Figure 16.5.2, summarises this process. Worked example 16.5.2 will help you to understand these steps.

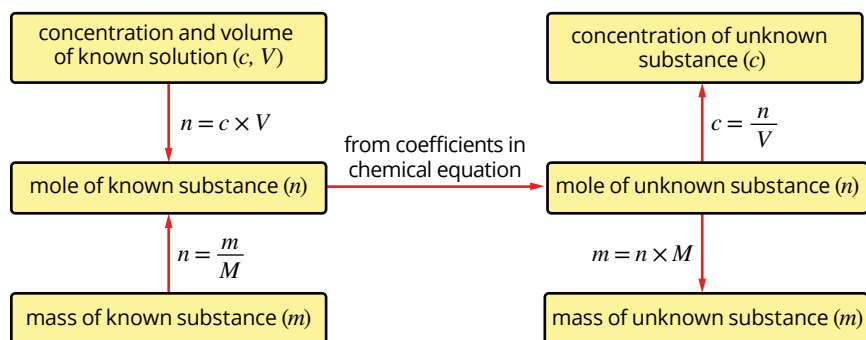


FIGURE 16.5.3 A flow chart for determining the quantity of a reactant or product by stoichiometric calculations is helpful when trying to solve these problems.

Worked example 16.5.2

STOICHIOMETRY PROBLEMS INVOLVING SOLUTIONS

Calculate the concentration of a sodium carbonate solution if 2.36 g of copper(II) carbonate is precipitated when 25.0 mL of the sodium carbonate solution is reacted with excess copper(II) sulfate solution.	
Thinking	Working
Write a balanced equation for the reaction.	$\text{Na}_2\text{CO}_3(\text{aq}) + \text{CuSO}_4(\text{aq}) \rightarrow \text{CuCO}_3(\text{s}) + \text{Na}_2\text{SO}_4(\text{aq})$
Calculate the number of moles of the known substance using $n = \frac{m}{M}$. The 'known substance' is the one you are provided information about in the question.	$M(\text{CuCO}_3) = 123.5 \text{ g mol}^{-1}$ $n(\text{CuCO}_3) = \frac{m(\text{CuCO}_3)}{M(\text{CuCO}_3)}$ $= \frac{2.36}{123.5}$ $= 0.0191 \text{ mol}$
Write the mole ratio for the known and unknown substances.	$\text{mole ratio} = \frac{n(\text{Na}_2\text{CO}_3)}{n(\text{CuCO}_3)} = \frac{1}{1}$
Calculate the number of moles of the unknown substance using: $n(\text{unknown}) = n(\text{known}) \times \text{mole ratio}$ The 'unknown substance' is the one whose mass you are required to calculate.	$n(\text{Na}_2\text{CO}_3) = 0.0191 \times \frac{1}{1}$ $= 0.0191 \text{ mol}$
Calculate the quantity (volume in this example) of the unknown substance by rearranging: $c = n \times V$	$c(\text{Na}_2\text{CO}_3) = \frac{n}{V}$ $= \frac{0.0191}{0.025}$ $= 0.764 \text{ mol L}^{-1}$

Worked example: Try yourself 16.5.2

STOICHIOMETRY PROBLEMS INVOLVING SOLUTIONS

Calculate the concentration of a potassium sulfate solution if 7.88 g of barium sulfate is precipitated when 55.0 mL of the potassium sulfate solution is reacted with excess barium nitrate solution.

16.5 Review

SUMMARY

- A balanced equation shows the ratio of the amount (in moles) of reactants and products formed in the reaction.
- Given the quantity of one of the reactants or products of a chemical reaction, such as in a precipitation reaction, the quantity of all other reactants and products can be predicted by working through the following steps:
 - Write a balanced equation for the reaction.
 - Calculate the amount (in moles) of the given substance.
 - Use the mole ratios of reactants and products in the balanced equation to calculate the amount (in moles) of the required substance.
 - Use the appropriate formula to determine the required answer. Formulae are:
$$n = \frac{m}{M}$$
$$n = c \times V$$

KEY QUESTIONS

- 1 Calculate the mass of precipitate produced when 120.0 mL of 0.075 mol L⁻¹ calcium chloride solution reacts completely with a sodium sulfide solution.
- 2 Calculate the volume of 0.050 mol L⁻¹ lead(II) nitrate solution that reacts with sodium iodide solution to produce 0.984 g of lead(II) iodide precipitate.
- 3 Silver nitrate solution will react with sodium chloride solution to give a precipitate of silver chloride according to the equation:
$$\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$$
Calculate the mass of silver chloride produced when 2.40 g of silver nitrate in aqueous solution is added to excess sodium chloride.
- 4 What volume of 0.125 mol L⁻¹ calcium nitrate solution will be needed to precipitate all the phosphate from 30.0 mL of a 0.155 mol L⁻¹ sodium phosphate solution? The balanced equation for the reaction is:
$$3\text{Ca}(\text{NO}_3)_2(\text{aq}) + 2\text{Na}_3\text{PO}_4(\text{aq}) \rightarrow \text{Ca}_3(\text{PO}_4)_2(\text{s}) + 6\text{NaNO}_3(\text{aq})$$

Chapter review

KEY TERMS

concentrated solution	ionic equation
concentration	molarity
dilute solution	parts per million (ppm)
dilution	precipitate
full equation	precipitation reaction

Precipitation reactions

- Consider the reaction represented by the following equation:
$$\text{NaCl(aq)} + \text{AgNO}_3\text{(aq)} \rightarrow \text{AgCl(s)} + \text{NaNO}_3\text{(aq)}$$
Indicate whether the following statements about the reaction and equation are true or false.
 - The equation represents a precipitation reaction.
 - Ionic compounds that contain sodium or nitrate ions are always soluble.
 - The precipitate in this reaction is AgNO_3 .
 - The equation is an example of an ionic equation.
 - Silver chloride is insoluble in water.
 - The sodium and chloride ions can be described as spectator ions.
- Copy and complete the following table. Identify which reaction mixtures will produce precipitates and write their formulae.

	NaOH	KBr	NaI	MgSO_4	BaCl_2
$\text{Pb(NO}_3)_2$					
KI					
CaCl_2					
Na_2CO_3					
Na_2S					

- What precipitate will be formed (if any) when the following solutions are mixed?
 - barium nitrate and sodium sulfate
 - sodium chloride and copper(II) sulfate
 - magnesium sulfate and lead(II) nitrate
 - potassium chloride and barium nitrate
- Write ionic equations for each of the following precipitation reactions.
 - $\text{NH}_4\text{Cl(aq)} + \text{AgNO}_3\text{(aq)} \rightarrow$
 - $\text{FeCl}_2\text{(aq)} + \text{Na}_2\text{S(aq)} \rightarrow$
 - $\text{Fe(NO}_3)_3\text{(aq)} + \text{KOH(aq)} \rightarrow$
 - $\text{CuSO}_4\text{(aq)} + \text{NaOH(aq)} \rightarrow$
 - $\text{Ba(NO}_3)_2\text{(aq)} + \text{Na}_2\text{SO}_4\text{(aq)} \rightarrow$

- Write an ionic equation for the reaction that takes place when solutions of the following compounds are mixed. In each case, give the spectator ions. Refer to the solubility tables on page xx to help identify the precipitate for each reaction.
 - copper(II) sulfate and sodium carbonate
 - silver nitrate and potassium chloride
 - sodium sulfide and lead(II) nitrate
 - iron(III) chloride and sodium phosphate
 - iron(III) sulfate and potassium hydroxide

Concentration of solutions

- What is the concentration, in ppm, of lead in a 6.0 kg solution that contains 12 mg of lead?
 - What is the concentration as percentage by mass?
- The label on a laundry bleach lists the concentration of the active ingredient, sodium hypochlorite, as 42 g/L (42 g L^{-1}). What mass of sodium hypochlorite will be in a 750 mL bottle of the bleach?

Molar concentration

- What is the molar concentration of a 2.0 L solution containing 30 g of NaOH?
- How many moles of solute are present in the following solutions?
 - 12 mL of 0.22 mol L^{-1} NaI
 - 150 mL of $0.0250 \text{ mol L}^{-1}$ KMnO_4
 - 7.2 L of $3.15 \times 10^{-3} \text{ mol L}^{-1}$ KBr
- What is the mass, in grams, of solute present in the following solutions?
 - 100 mL of 1.20 mol L^{-1} NH_3
 - 20 mL of 0.50 mol L^{-1} AgNO_3

Dilution

- How much water must be added to 25 mL of a 4.0 mol L^{-1} solution of potassium carbonate solution to dilute it to a concentration of 1.6 mol L^{-1} ?
- 1.50 mL of a $0.0500 \text{ mol L}^{-1}$ solution of calcium chloride (CaCl_2) was diluted with water to a volume of 10.0 L. What was the concentration of Cl^- ions in the diluted solution in ppm? Assume 1.0 L of the final solution has a mass of 1.0 kg.

Calculations involving reactions in solutions

- 13** What mass of precipitate is produced when 25.6 mL of 0.550 mol L^{-1} copper(II) chloride solution reacts with excess potassium carbonate solution?
- 14** What volume of 0.752 mol L^{-1} sodium iodide solution is needed to precipitate all the lead(II) ions from 150.0 mL of 0.085 mol L^{-1} lead(II) nitrate solution?

- 16** 100.0 mL of 0.150 mol L^{-1} sodium sulfate solution reacts completely with 25.0 mL of barium nitrate solution.

- a** What mass of precipitate is produced from the reaction?
- b** What is the molar concentration of the sodium ions after reaction?

Connecting the main ideas

- 15** Copy and complete the diagram below by inserting into the white boxes the processes required to convert between the quantities in the yellow boxes. The answer to the top left box is given to you.

